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India's semiconductor ambitions are evolving beyond manufacturing. The next phase may be defined by ecosystem depth, innovation and talent.

- **Semiconductor manufacturing in India enters a new phase**
- **Building India's semiconductor ecosystem**

- **Semicon 2.0 broadens India's semiconductor strategy**
 - **India's strengths extend beyond incentives**
 - **The next phase may be defined by execution**
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In brief

- India's semiconductor ambitions are translating into results, with 12 approved projects, 23 chip tape-outs, and facilities entering production and exports.
 - India's next phase of growth may depend less on new fabs and more on building capabilities in materials, equipment, advanced packaging, R&D and talent.
 - Semicon 2.0 shifts the focus from attracting investments to building capabilities across design, manufacturing, materials, advanced packaging, R&D and talent.
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India's semiconductor ambitions are entering a new phase. As semiconductors become increasingly central to economic competitiveness, technological sovereignty and supply-chain resilience, countries around the world are seeking to strengthen domestic capabilities across the semiconductor value chain. For India, this opportunity is being shaped by a growing domestic market, established design strengths and an evolving policy framework led by the India Semiconductor Mission and Semicon 2.0.

The country has already made significant progress in attracting investments into semiconductor fabrication, packaging and related manufacturing activities. However, global experience shows that long-term competitiveness is built not by individual manufacturing projects alone, but by the depth of the ecosystem that supports them. This is increasingly reflected in India's semiconductor strategy, which is expanding beyond manufacturing to encompass design, research, advanced packaging, materials, equipment and talent development.

Semiconductor manufacturing in India enters a new phase

For decades, India's participation in the semiconductor industry was defined primarily by engineering and design talent. While the country emerged as a leading destination for chip design and R&D activities, manufacturing capabilities remained limited.

That equation is beginning to change. India's semiconductor market has grown from approximately US\$27 billion in 2021 to nearly US\$64 billion in 2026 and is projected to reach US\$200 billion by 2035. In response, the Government of India launched the India Semiconductor Mission and introduced targeted interventions to support domestic manufacturing.

Under Semicon 1.0, twelve manufacturing projects across fabrication, compound semiconductors and packaging have been approved, attracting investments of approximately INR1.64 lakh crore. Several facilities have already commenced production and exports, marking an important step in the growth of semiconductor manufacturing and the broader semiconductor value chain in India.

Yet manufacturing capacity alone is unlikely to determine long-term success.

Building India's semiconductor ecosystem

Leading semiconductor economies have demonstrated that global competitiveness depends on the strength of an interconnected ecosystem. Fabrication facilities rely on sophisticated networks of materials suppliers, equipment manufacturers, advanced packaging providers, research institutions and highly specialized talent.

As semiconductor technologies become more complex, innovation is increasingly driven by areas such as advanced packaging, chiplets, heterogeneous integration, silicon photonics and system-level design. Consequently, countries such as the United States, Japan, South Korea, Taiwan and members of the European Union are supporting not only manufacturing, but also research, commercialization, talent development and supply-chain capabilities.

Recognizing this reality, India's semiconductor strategy is increasingly focused on building ecosystem depth around emerging manufacturing investments.

Semicon 2.0 broadens India's semiconductor strategy

The evolution from Semicon 1.0 to Semicon 2.0 reflects this shift in policy thinking.

While Semicon 1.0 focused on establishing foundational capabilities through fiscal support of up to 50% of capital expenditure, Semicon 2.0 adopts a broader six-pillar approach covering design, machines and materials, fabrication, advanced packaging, R&D and talent development.

Importantly, the program introduces differentiated support aligned with strategic priorities. Projects related to research and talent development may receive support of up to 75% of total project costs when combined with state incentives, reflecting the growing importance of capability creation in areas that typically require long development cycles.

The program also expands support beyond traditional capital subsidies. Semiconductor startups and design companies can access mechanisms such as seed funding, equity co-investment and royalty-linked financing, helping bridge the gap between innovation and commercialization.

India's strengths extend beyond incentives

While incentives remain important, India's semiconductor proposition is built on a wider set of advantages.

India is estimated to host nearly 20% of the world's semiconductor design talent and continues to strengthen its position in semiconductor design through initiatives such as the Design Linked Incentive (DLI) Scheme and Chips-to-Startup (C2S) program. Growing domestic demand, expanding electronics manufacturing, a large engineering workforce and increasing adoption of digital technologies further support long-term semiconductor growth.

India's policy framework also includes a distinctive milestone-linked pari passu incentive mechanism. Unlike models that rely primarily on tax credits or retrospective reimbursements, support under the India Semiconductor Mission is disbursed alongside project implementation and investment commitments, improving project visibility and supporting execution in a highly capital-intensive industry.

The next phase may be defined by execution

India has established the foundations of a semiconductor ecosystem, but sustaining momentum will require continued execution. Strengthening supplier ecosystems, expanding semiconductor R&D capabilities, developing specialized manufacturing talent and accelerating capabilities in materials, equipment and advanced packaging will be critical.

State governments also have an important role to play in aligning policy support with the broader objectives of Semicon 2.0 and extending incentives to a larger portion of the value chain.

The opportunity is significant. With a projected semiconductor market of US\$200 billion by 2035, growing manufacturing investments, world-class design talent and an increasingly comprehensive policy framework, India possesses many of the ingredients required to become a leading semiconductor ecosystem.

The next phase of India's semiconductor journey may be defined not by the number of fabs announced, but by the ecosystem built around them. The real opportunity lies in translating manufacturing momentum into ecosystem leadership.

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Learn about India's semiconductor ecosystem, Semicon 2.0 and investment opportunities.

1. Why are semiconductors important for India's economic growth and semiconductor strategy?

Semiconductors are important for India's economic growth because they power industries such as electronics, **telecommunications**, automotive, **artificial intelligence**, cloud infrastructure, advanced manufacturing, healthcare and defence. As part of India's semiconductor strategy, building a strong India semiconductor ecosystem can support industrial competitiveness, attract semiconductor investments in India, strengthen technological resilience and reduce concentration risks across strategically important global **supply chains**. Expanding semiconductor manufacturing in India can also drive innovation, employment and economic value creation.

2. What incentives are available under the India Semiconductor Mission and Semicon India 2.0?

The India Semiconductor Mission provides financial support for developing India's semiconductor ecosystem. Semicon India 1.0 offered fiscal support of up to 50% of capital expenditure. Under Semicon 2.0, support is differentiated by activity across the semiconductor value chain in India, including design, fabrication, packaging, R&D and talent development. When combined with state incentives, certain research and talent projects may receive support covering up to 75% of project costs.

3. How is Semicon 2.0 different from Semicon 1.0?

Semicon 1.0 established foundational capabilities in semiconductor design in India, fabrication and packaging, with fiscal support of up to 50% of capital expenditure. Semicon 2.0 broadens this approach by supporting six interconnected pillars of India's semiconductor ecosystem and introducing differentiated incentives, financing mechanisms and greater emphasis on semiconductor R&D in India, advanced packaging, innovation and talent development. The program reflects the next phase of India's semiconductor policy.

4. How does Semicon 2.0 support semiconductor startups and MSMEs?

Semicon 2.0 supports startups and companies involved in semiconductor design in India through seed funding and equity co-investment financing structures. These mechanisms are intended to bridge financing and commercialization gaps between chip development and market deployment. MSMEs operating across the semiconductor value chain in India, including materials, equipment and related services, may also benefit from the program's broader ecosystem-development focus.

5. What opportunities does India offer semiconductor manufacturers and investors?

India offers semiconductor manufacturers and investors a growing domestic electronics market, a globally significant design base, an expanding manufacturing footprint and policy support under the India Semiconductor Mission. Opportunities span chip manufacturing, packaging, materials, equipment and R&D. Milestone-linked incentives, state-level support and demand from technology-intensive industries further strengthen the outlook for semiconductor investments in India.

6. How can India become a global semiconductor hub?

India can become a global semiconductor hub by converting manufacturing momentum into a connected and competitive semiconductor ecosystem. This requires stronger semiconductor clusters, domestic suppliers, materials and equipment capabilities, advanced semiconductor packaging in India, reliable infrastructure, specialized talent, research commercialization and industry-academia collaboration. Enhanced fiscal support and better alignment between state policies and Semicon 2.0 objectives can further accelerate semiconductor ecosystem development in India and support the India Semiconductor Roadmap 2035.

7. What are the key challenges facing India's semiconductor ecosystem?

India's semiconductor ecosystem faces gaps in funding scale, materials and equipment manufacturing, advanced packaging, research infrastructure, specialized talent and critical utilities. Several state policies also remain concentrated on semiconductor fabrication and packaging. Addressing these challenges will require stronger supplier networks, reliable ultra-pure water and utilities, industry-aligned training, technology commercialization and broader state-level semiconductor support to strengthen the semiconductor supply chain in India.

Summary

India's semiconductor journey is entering a new phase. Having established early manufacturing momentum through strategic investments and policy support, the focus is now shifting toward ecosystem development. Strengthening materials, equipment, advanced packaging, research, talent and innovation capabilities will be critical to India's long-term competitiveness. With growing domestic demand, strong design capabilities and an expanding policy framework, India has an opportunity to emerge as a trusted and globally competitive semiconductor ecosystem.

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